Using a seismograph 6a

Earthquakes are often measured using the Richter scale. The Richter scale is better known than the Mercalli scale and is used with seismographs. A seismograph is a device that measures the strength of earthquakes.

Building a seismograph

The other day we built and tested a seismograph in our science class. We hung a 2 kilogram weight from a retort stand. The retort stand had a clamp fixed to its top with a boss head. A strong string held the weight way below the clamp. The weight was about 5 centimetres above the base of the retort stand. Then we taped a pen to the weight so that the pen just touched the retort stand base. Finally we slipped a sheet of paper between the pen and retort stand base. And that was our seismograph.

1 Make a labelled sketch of the seismograph described above.

The teacher then told us to simulate an earthquake by gently shaking the table for a few seconds. At the same time one of us had to pull the paper slowly across the retort stand base to make a seismogram.

2 Define the terms 'seismograph' and 'seismogram'.

3 Complete the seismogram by continuing to draw the line during the simulated earthquake until just after it finishes.



4 Describe the line on the seismogram when there is no earthquake and also when there is an earthquake.

5 Describe the line on a seismogram during a strong earthquake.

It was a great activity, trying to imitate an earthquake. We then repeated the simulation but this time we had a really big earthquake. When we stopped shaking the table, we noticed that the weight was swinging like a pendulum. Our seismogram looked a little strange as well.

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Using a seismograph

6b

6 Draw a possible seismogram line for the situation described above.



before during after the earthquake

7 Explain why the weight was moving after the big earthquake.

8 Explain how a seismograph works to make a seismogram, using the scaffold below.

Title	
Purpose	
Explanation sequence	
Example	When the Newcastle earthquake occurred in December 1989, it measured 5.6 on the Richter scale. The earthquake lasted only 5 seconds but was felt 5000 km away. Ten people died, 160 people were hospitalised and 50 000 houses were damaged. An aftershock earthquake three days later produced 2.1 on the Richter scale.

9 Suggest how the school seismograph could be improved.

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